

Appendix A — Full Roundtrip Table (per-role breakdown)

The met-

ric used throughout the paper is the `role_match_score` returned by `cogant.reverse.idempotency.compute_isomorphism_report(orig_synth_gnn)`. It is a multiset similarity over the role populations of the forward GNN (`orig_gnn`) and the re-forwarded synthesized package (`synth_gnn`). In this appendix we further decompose into four per-role components `_HIDDEN_STATE`, `_OBSERVATION`, `_ACTION`, and `_CONSTRAINT`, each computed as the multiset-similarity restricted to a single role category:

$$_role(P) = \min(count_orig(role), count_synth(role)) / \max(count_orig(role), count_synth(role))$$

with the convention `_role = 1.0` when both counts are zero (the role is vacuously preserved) and `_role = 0.0` when exactly one of the two counts is zero (the role has been introduced or dropped). The overall `_role` reported by `compute_isomorphism_report` is the mean of the per-role components over the roles present in at least one side, which matches the values reported in

